

HACEMOS FUTURO

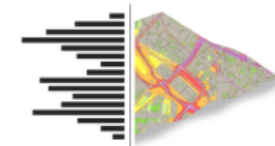
Subdirección General de
Prevención de la
Contaminación

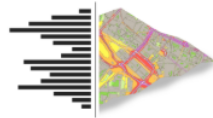


National environmental noise assessment

Methodologies for nationwide quantitative assessment

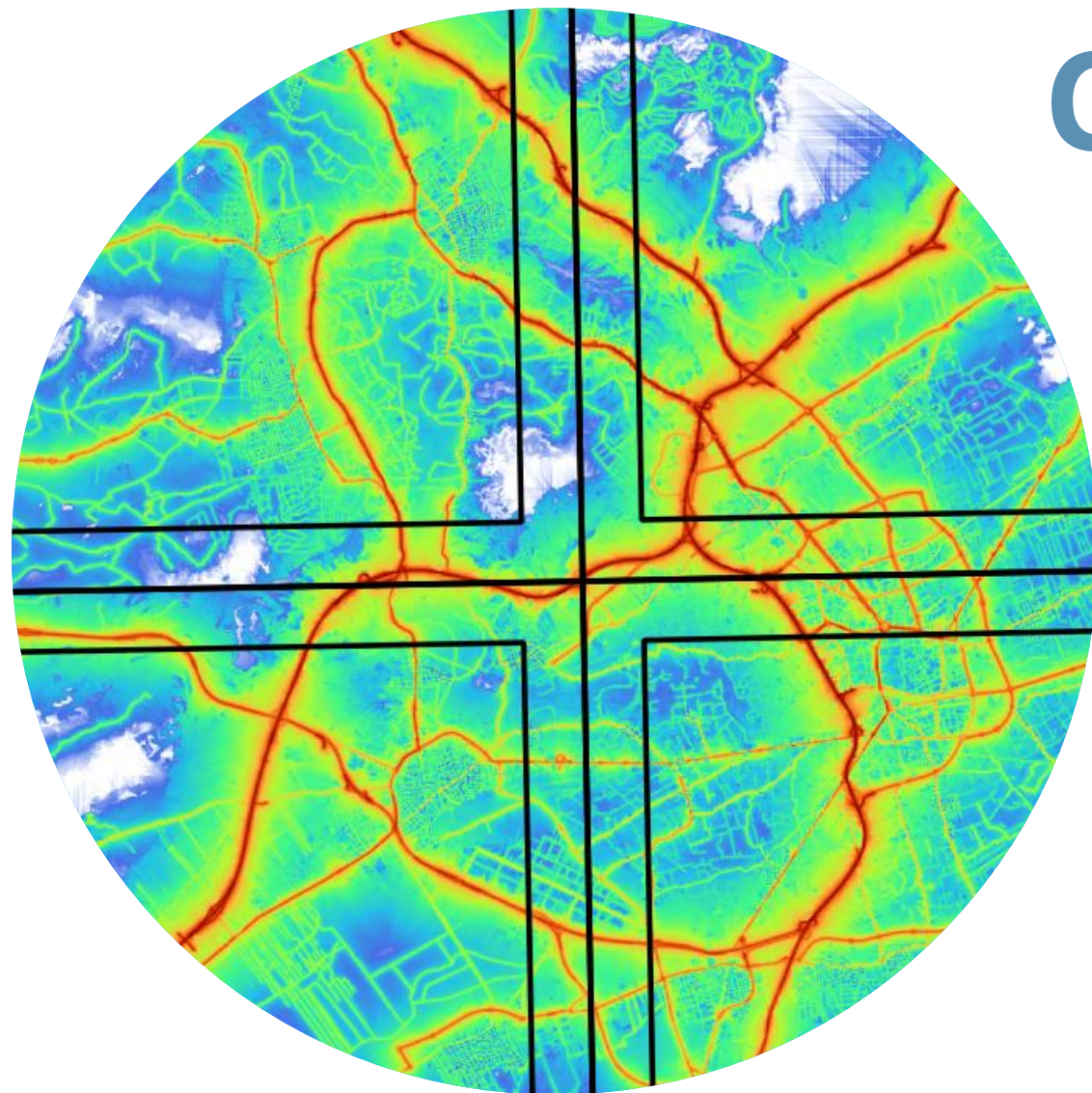
NoiseModelling Days 2026

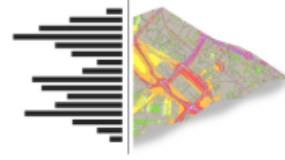




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- 2 Common methodology
- 3 Road, Railway and wind turb.
- 4 Airports
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01



Need for an assessment

Assessment of all modelable sources

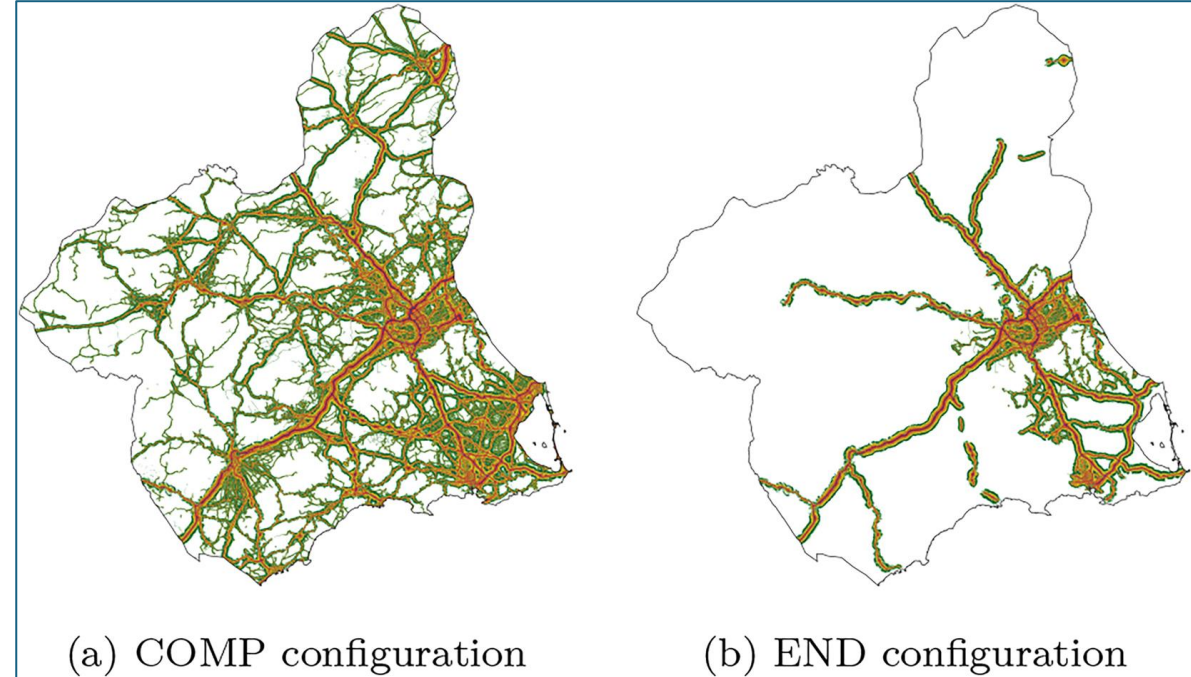
Why a national assessment is needed

The END framework focuses on:

- **Agglomerations** more than **100,000** inh.
- **Roads** with more than **3 million** vehicles/year.
- **Railways** more than **30,000** movements/year.
- **Airports** with more than **50,000** oper./year.

This leaves out a significant share of actual exposure.

- Municipalities not included in agglomerations.
- Infrastructure below END thresholds.
- Dispersed or non-traditional sources.
- Large rural and peri-urban areas.
- Some sensitive receptors in the natural environment.



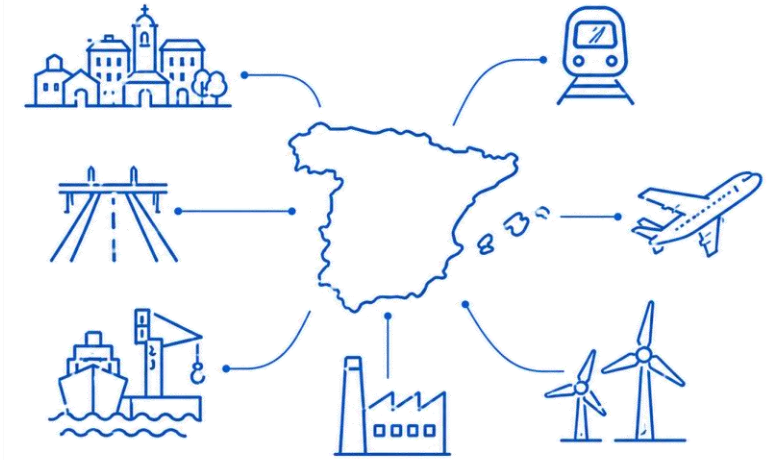
Sources outside the Directive focus

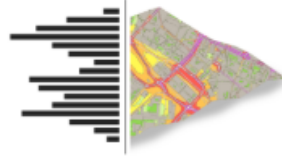
Population

- 87 municipalities with 50,000 to 100,000 inhabitants.
- 613 municipalities with 10,000 to 50,000 inhabitants.
- 2,381 municipalities with 1,000 to 10,000 inhabitants.
- 4,986 municipalities with fewer than 1,000 inhabitants.

Infrastructure

- 154,000 km of roads.
- 21,770 km of railway lines.
- 70+ relevant aeronautical facilities
- 9,067 industrial facilities registered in PRTR.
- 1,345 wind farms with more than 22,000 wind turbines.
- 49 state-owned port facilities.
- Local sources such as leisure activities, neighbourhood activities, urban services, recreational events and construction works.





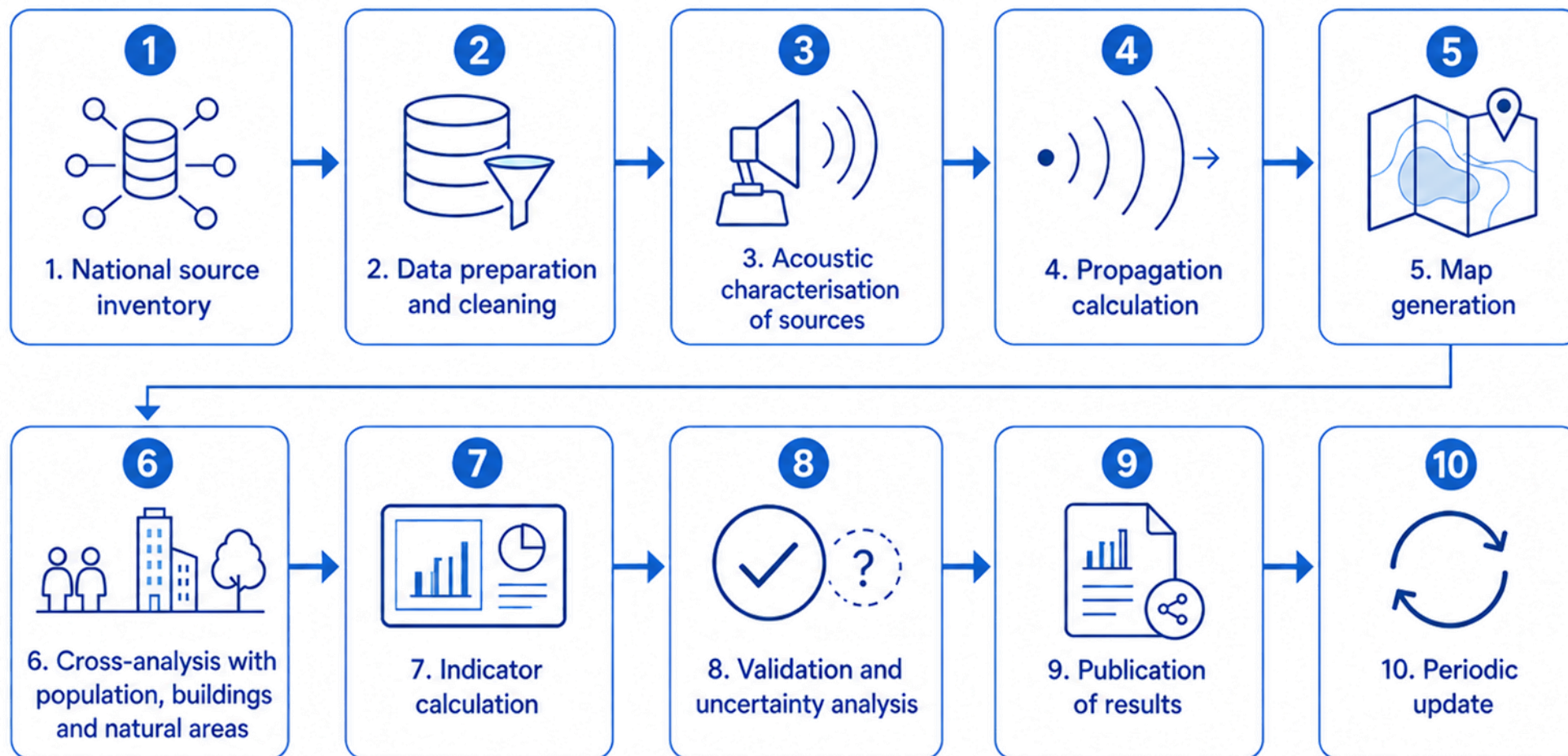
02



Common methodology

Consistent treatment of sources

Common assessment architecture



Role of supercomputing

Supercomputing is the element that **enables the assessment to scale**. It allows:

- Run **millions of source-receiver paths**.
- Work with **high-resolution grids**.
- Process **entire territories**.
- Split calculations by map sheet.
- **Parallelise** independent jobs.
- **Reduce execution times** from months to days.
- **Repeat calculations** when data are updated.



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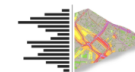
Nationwide High-Resolution Environmental Noise Modeling in Spain

AECT-2025-3-0005

Nationwide High-Resolution Environmental Noise Modeling in Spain

AECT-2025-3-0008

Dynamic Spatio-Temporal Noise Assessment from Aircraft in Spain

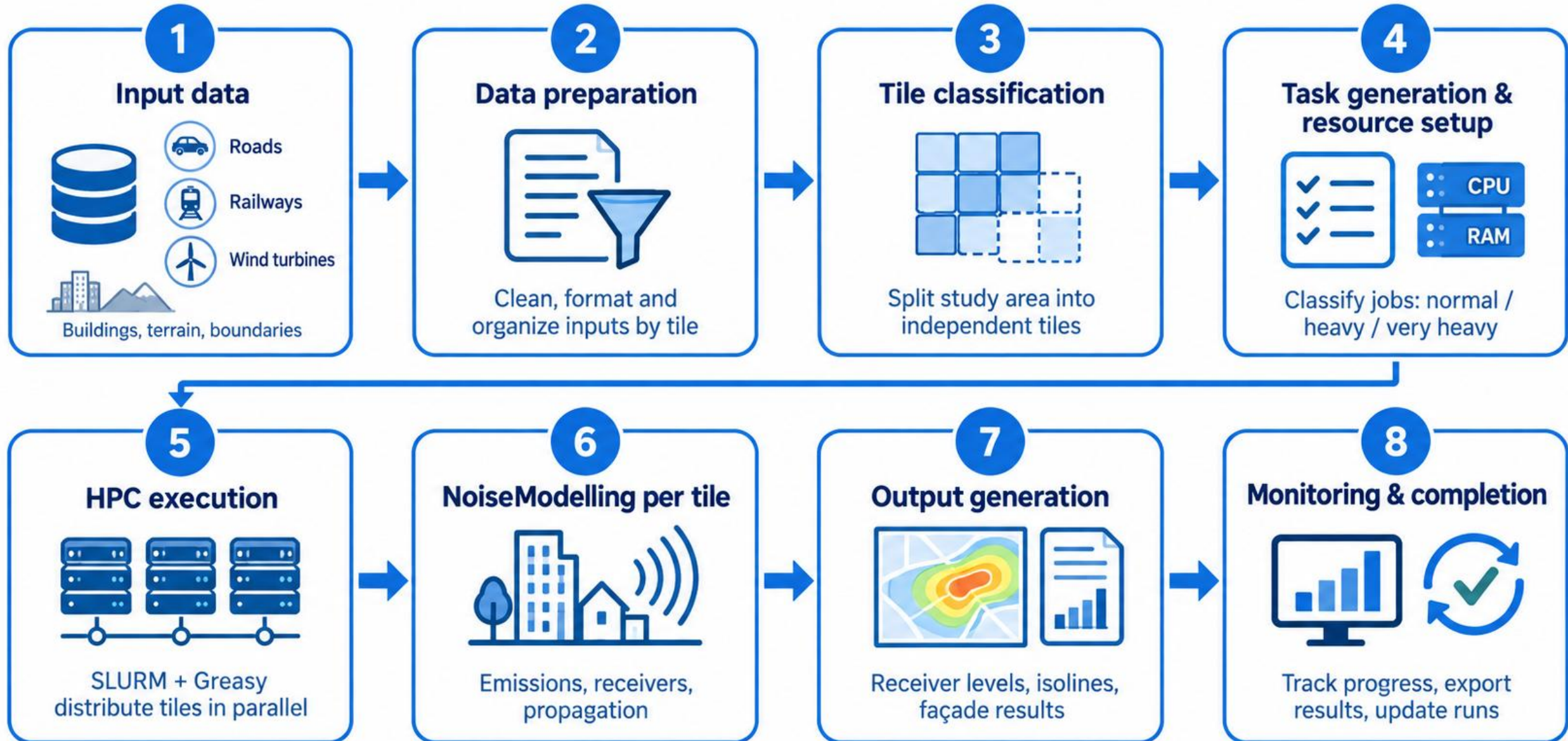


Simplified NoiseModelling Workflow on HPC

NoiseModelling Days 2026



Same workflow for roads, railways and wind turbines



Execution on supercomputing infrastructure

Job type	CPUs / task	JVM heap (Xms–Xmx)	Metaspace	Direct memory	Tasks per 112-core node*
Normal	6	4–8 GB	512 MB	512 MB	18
Heavy	10	6–12 GB	768 MB	768 MB	11
Very heavy	12	8–16 GB	1024 MB	1024 MB	9

*The scripts assume **112 CPU cores per node** to estimate capacity.

HPC resource configuration

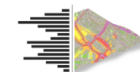
Available workflow:

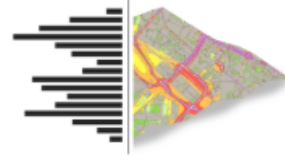
<https://github.com/IsotoCedex/NoiseModellingSupercomputerRoads>

SLURM configuration: Jobs are submitted with `--ntasks` and `--cpus-per-task`; for the normal, heavy and very-heavy task files, the current launcher does **not explicitly fix --nodes**, so node allocation is handled by SLURM.

Parallel execution: normal tiles are grouped into files of **9 tasks**, while heavy and very-heavy tiles are launched individually.

Per-task runtime: Java OpenJDK 11 is used; `OMP_NUM_THREADS` and `MKL_NUM_THREADS` are set equal to the allocated CPU count. The JVM uses **G1GC**, with `ParallelGCThreads = CPUs/task`.

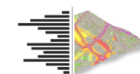
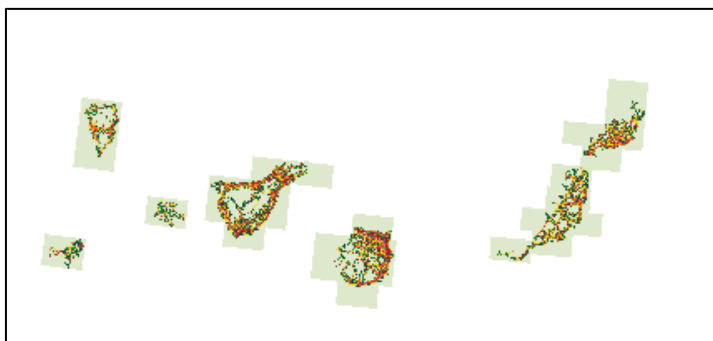
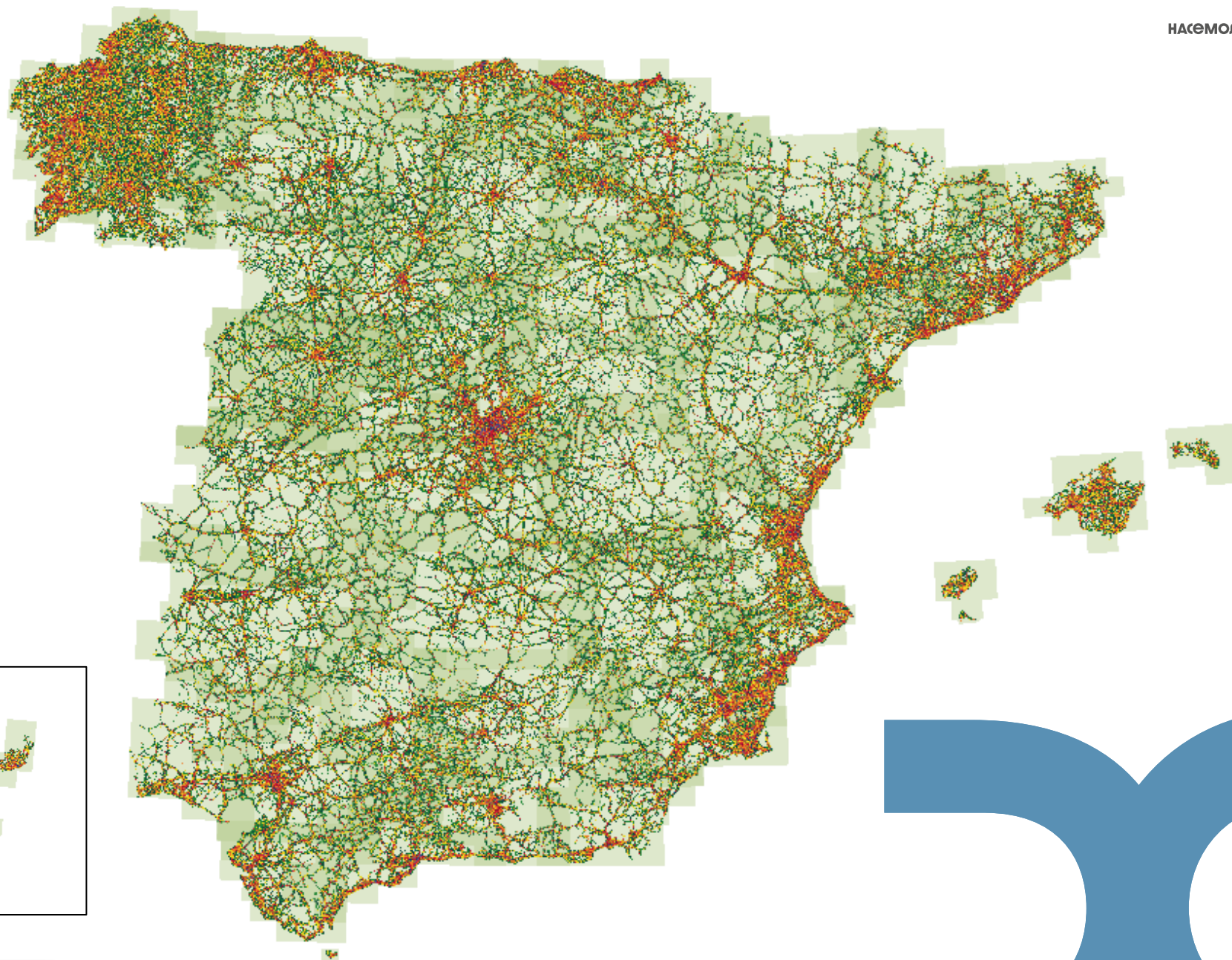


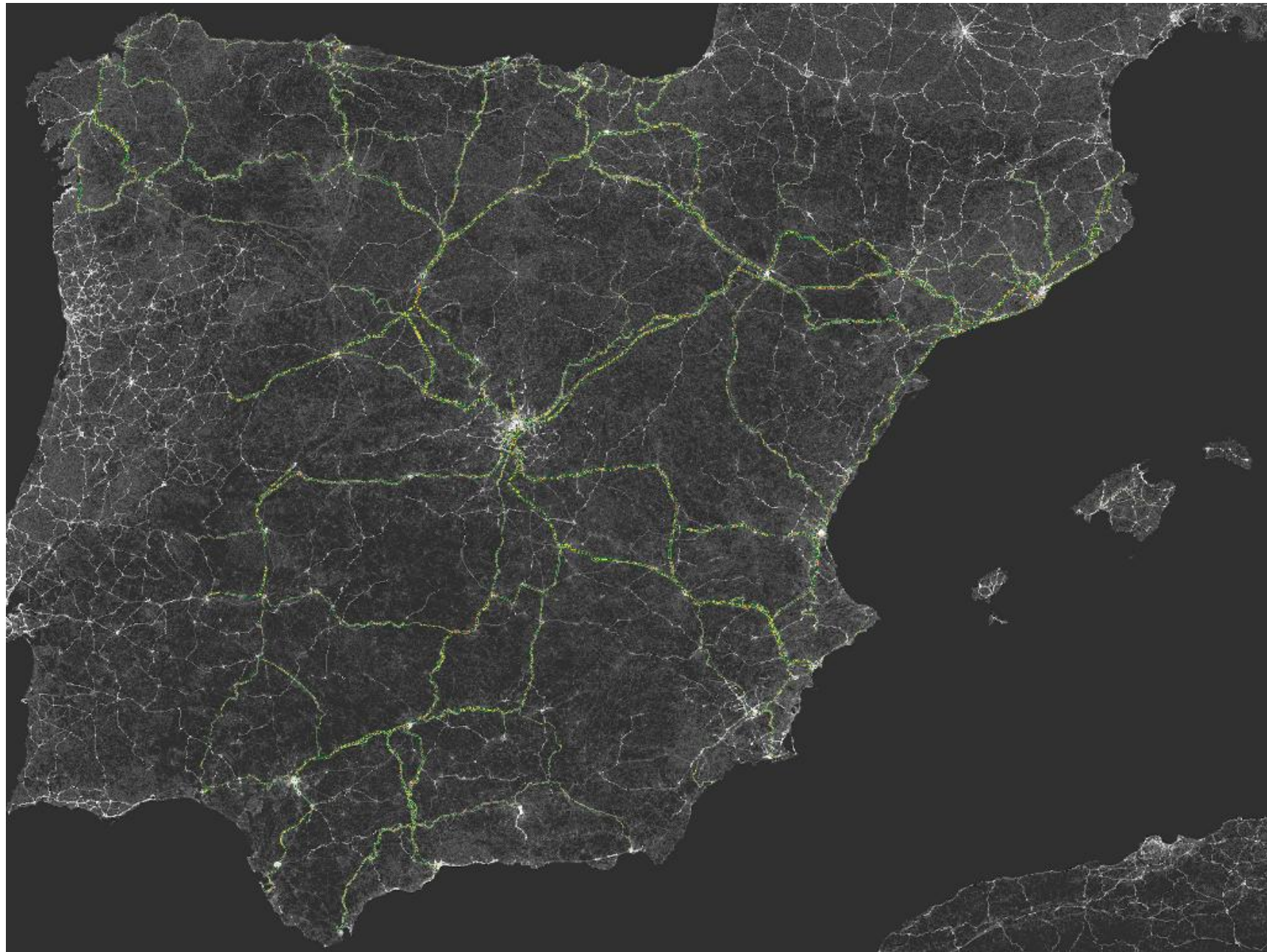


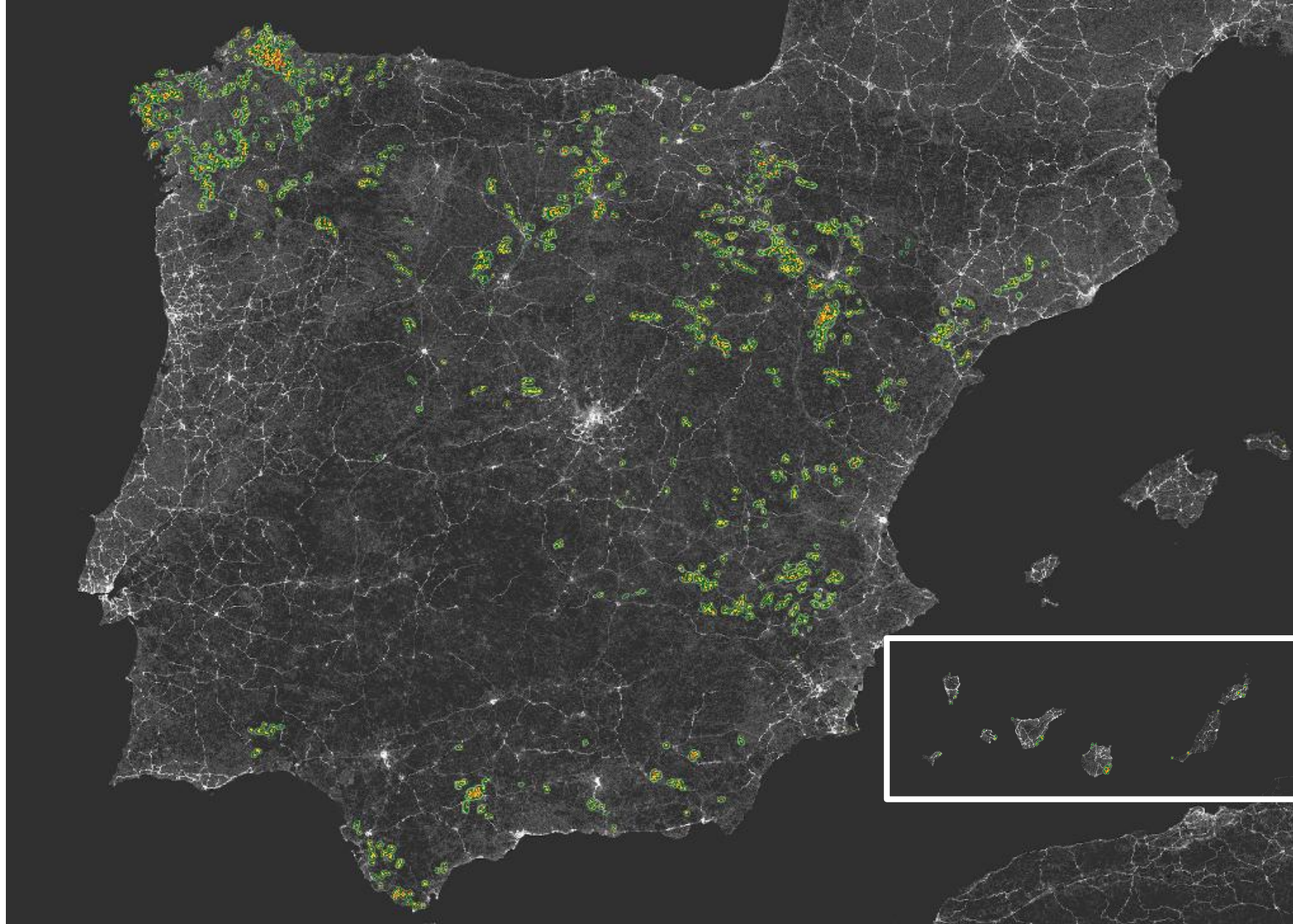
03

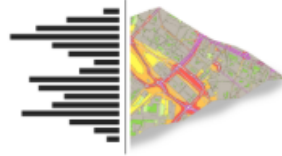


Roads, Railways and wind turbines









04

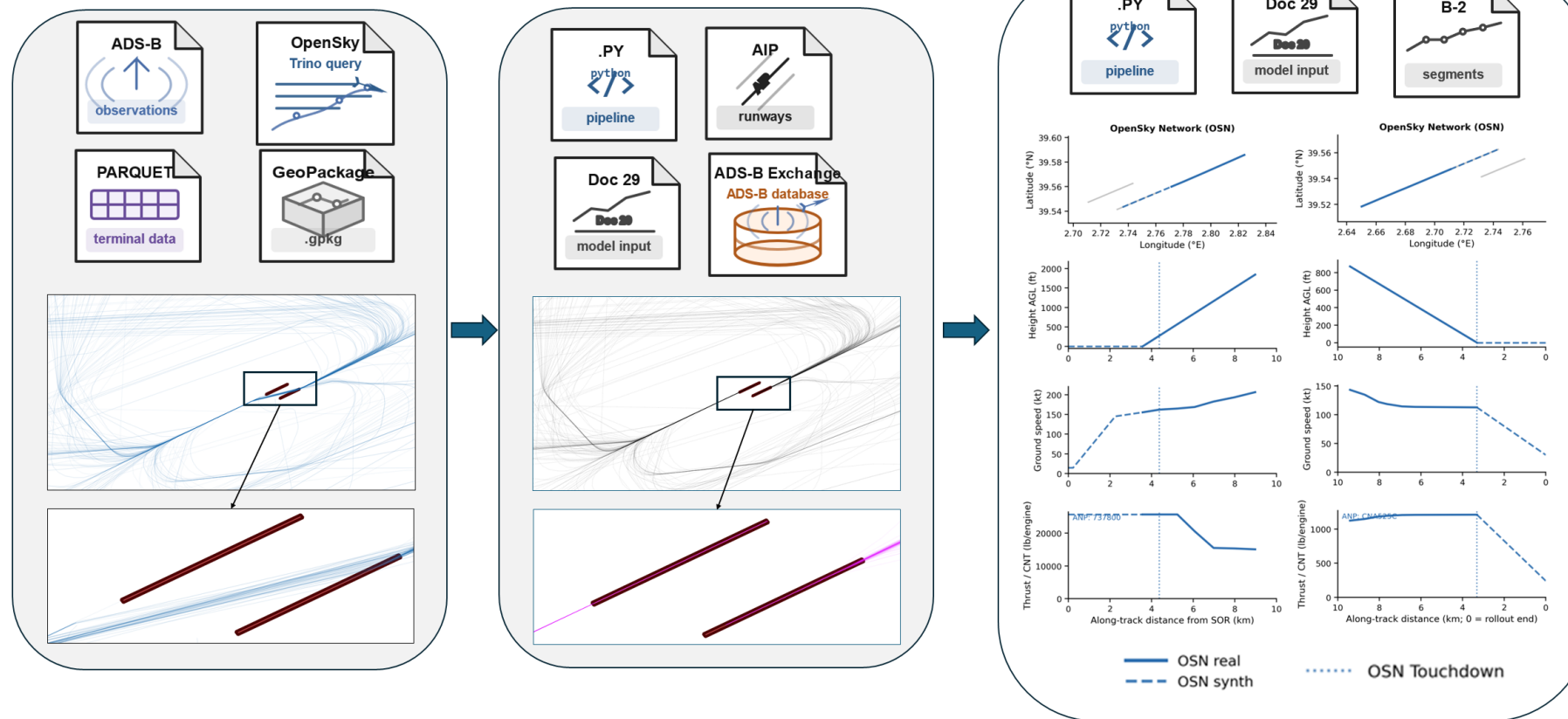


Airport model

61 airports modelled

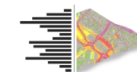
Aircraft methodology: trajectory preparation

Open ADS-B processing chain for ECAC Doc 29



Preprint available:

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7391664



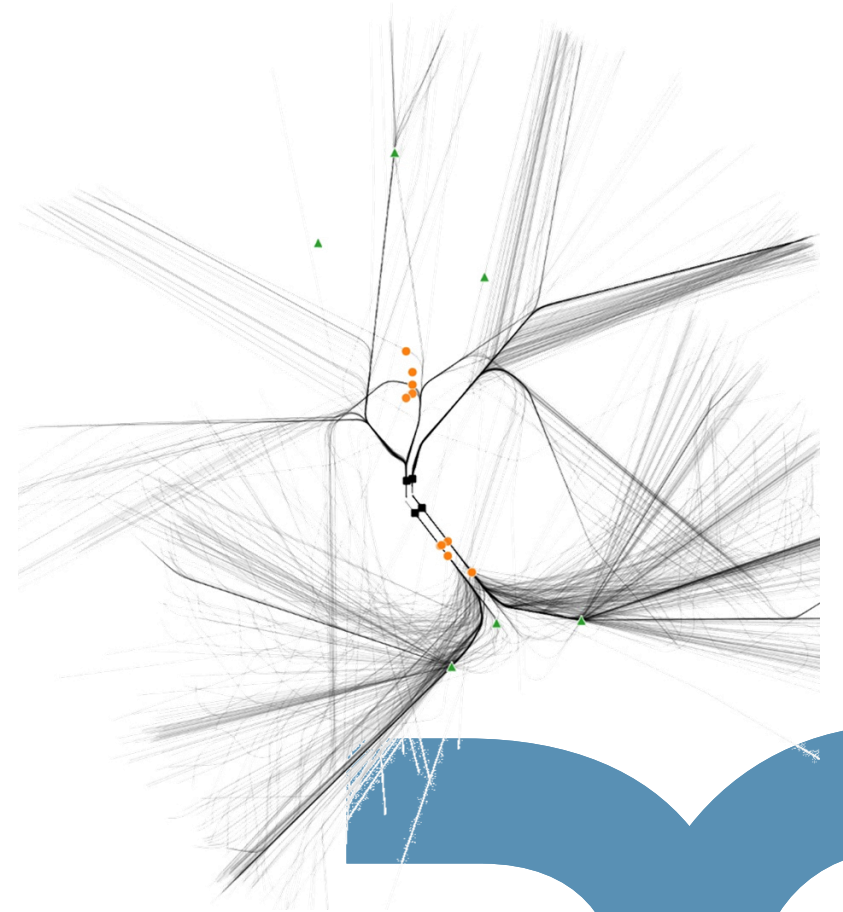
Aircraft methodology: aircraft characterisation

Each operation must be associated with an acoustically modelable aircraft.

Process:

- Aircraft type identification.
- Mapping to the ANP database (Aircraft Noise and Performance (ANP) - EASA - European Union).
- Substitutions when no direct match exists.
- Assignment of performance parameters.
- Calculation of speed, altitude and thrust by segment.
- Trajectory consistency check.

The goal is to produce standardised inputs for acoustic calculation.

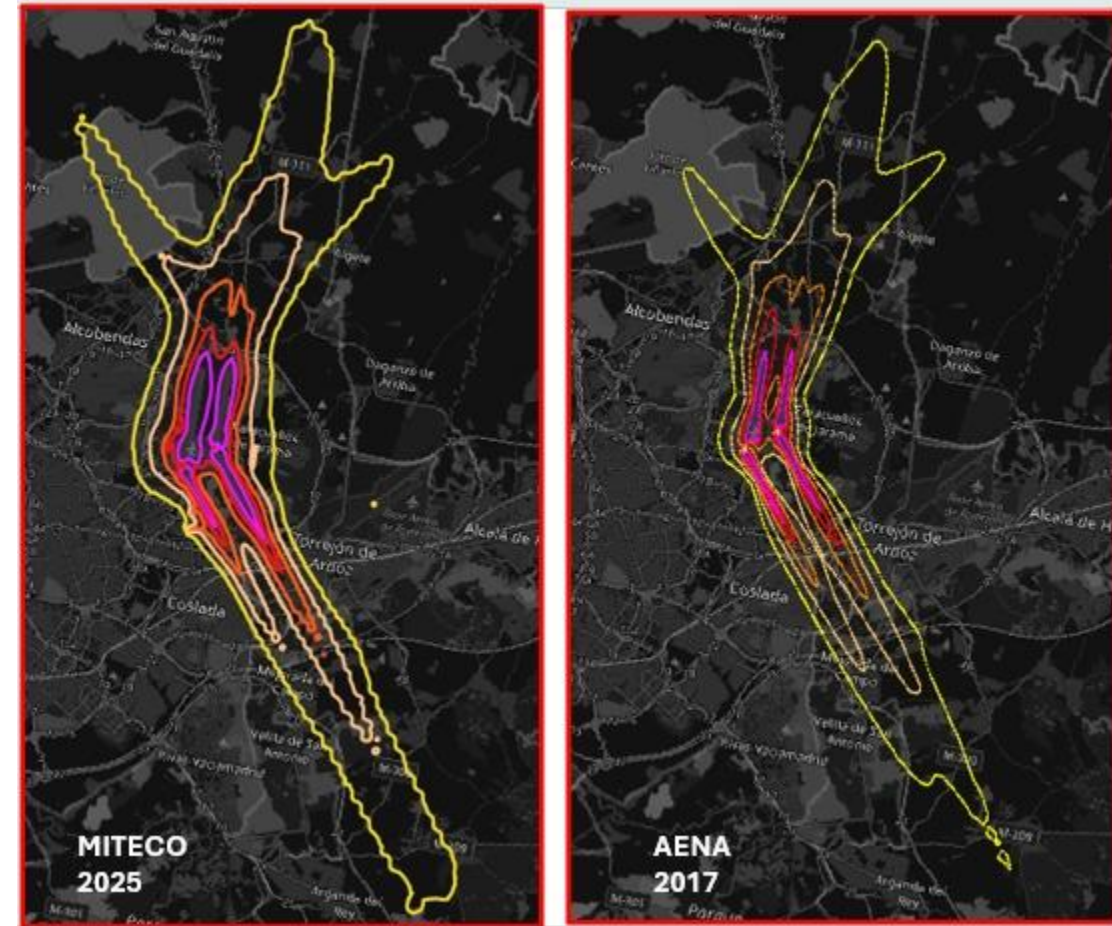


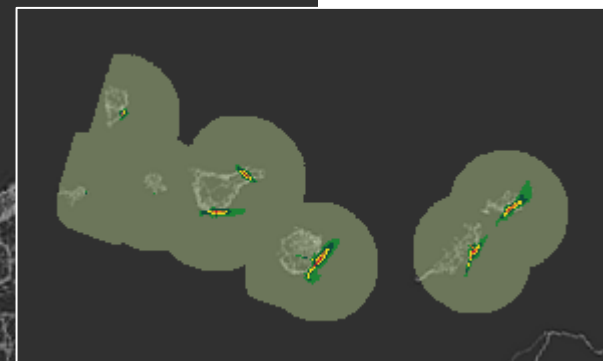
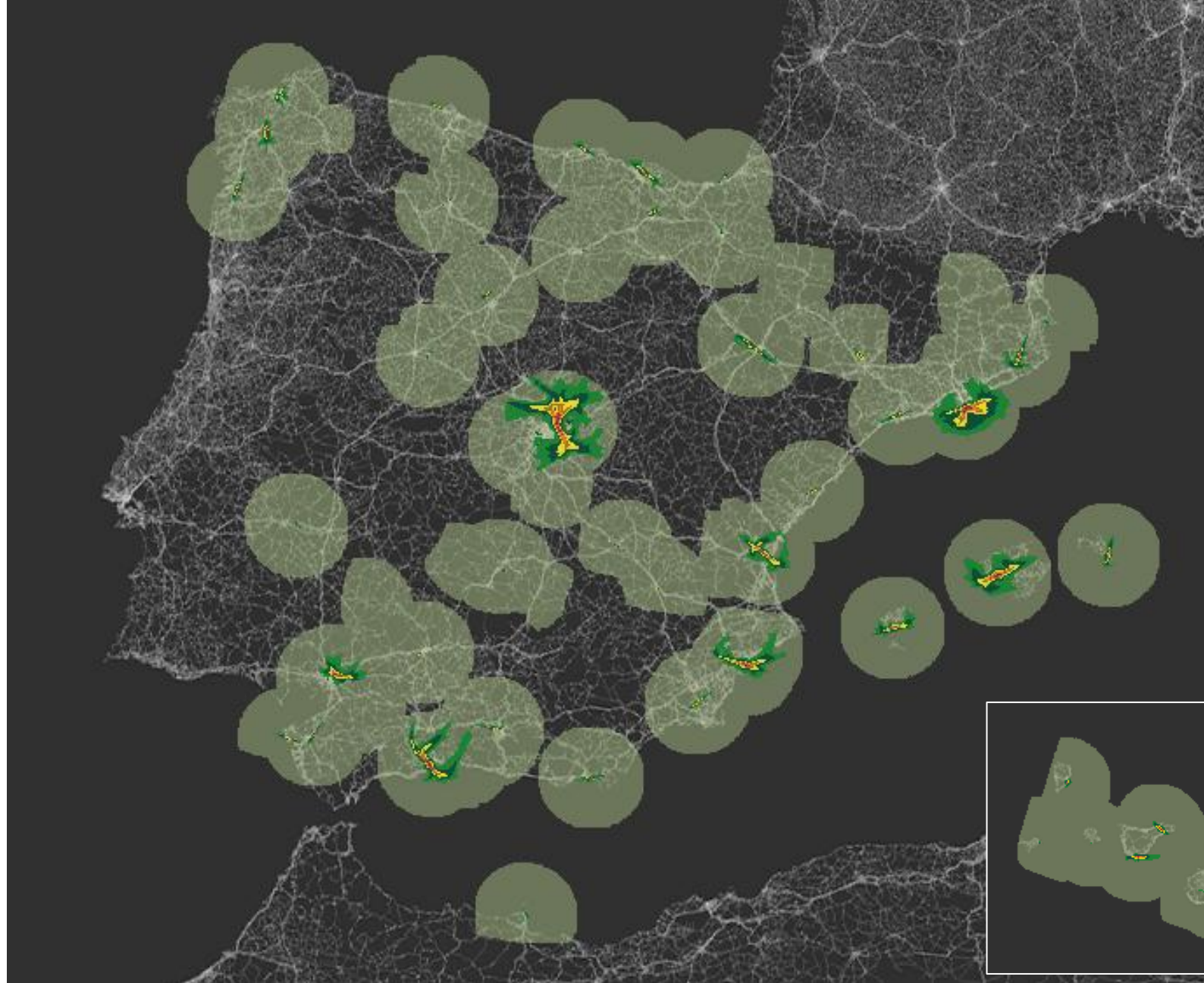
Aircraft methodology: ECAC Doc 29 calculation and integration

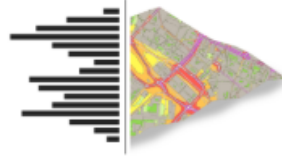
The acoustic calculation is based on ECAC Doc 29:

- Single-event noise.
- Cumulative levels.
- Lden and Lnight indicators.
- Contour generation.
- Receiver calculations.
- Integration by day, evening and night periods.

National execution requires automation, parallelisation by facility and quality control of modelled operations.







05



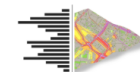
Conclusions

Conclusion

The quantitative assessment of noise at national level is technically feasible.

The key is to combine:

- Standardised methods → CNOSSOS, ECAC, ISO, National Methods
- Rigorous data preparation → Data Base
- Reproducible software → Noise Modelling
- Supercomputing → HPC



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Subdirección General de
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THANK YOU

Noise Unit

NoiseModelling Days 2026

